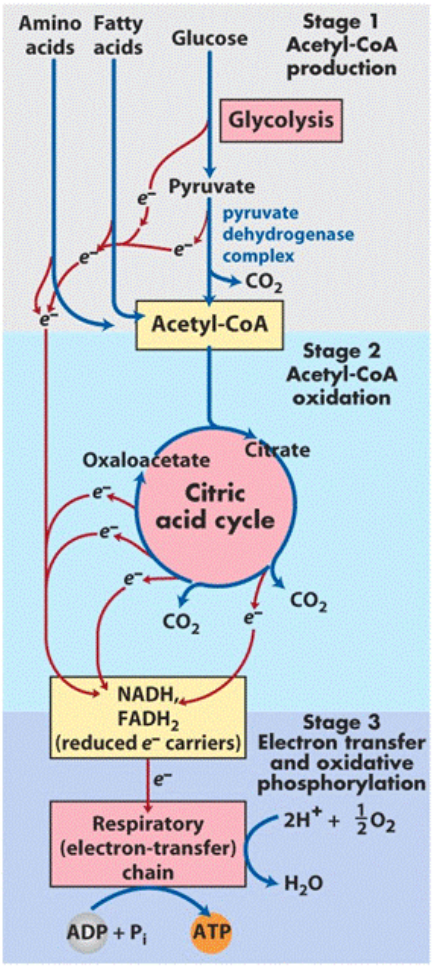
The Citric Acid Cycle (Krebs Cycle)

- The oxidation of glucose to pyruvate is a necessary step on the way to energy production, but not a sufficient one
- If you remember that process (glycolysis) you realize that the amount of ATP generated in it is 2 molecules.
- However both, the pyruvate produced in the process and the NADH that is generated, have a high potential to generate ATP through aerobic oxidative processes.
- Now we will see how pyruvate, instead of being reduced to lactate (fermentation), undergoes a total oxidation to CO_2
- This process, as a whole, leading to the complete oxidation of the organic fuels (glucose, fatty acids, and amino acids) is known as **cellular respiration**

- **Cellular respiration takes place in three stages**:
- **First stage**: carbon skeletons of organic fuels are oxidized to a common compound **called acetyl-coenzyme A.** Depending on the starting molecule, the route used to convert the initial carbons into acetate units is different. As we have already seen, **in the case of glucose, we will reach pyruvate through glycolysis**, and as we will see later, **pyruvate will be converted into acetate through an additional oxidative decarboxylation** reaction (pyruvate dehydrogenase reaction)



- **Second stage**: in this stage, the acetate (acetyl-CoA) is incorporated into **the citric acid cycle** (or Krebs cycle) where the two carbons are oxidized to CO_2. At this stage of cellular respiration, **most of the energy released during oxidations is temporarily conserved as NADH and $FADH_2$.**

- **Third stage**: the reduced cofactors (NADH and $FADH_2$) generated both in the first and in the second phase, are oxidized and the released electrons will end up reducing O_2, previously passing through a chain of electron-transport molecules. . In this process, **known as oxidative phosphorylation,** a large amount of energy is released that allows ATP to be synthesized

- As seen in the diagram, the Krebs cycle is the central axis of cellular metabolism
- It is the gateway to the aerobic metabolism for those molecules that will allow to produce energy
- It is also a pathway that provides precursors required for biosynthesis of other molecules, Aa, nucleotides, porphyrins, and even glucose (from oxaloacetate)
- Itself, the Krebs cycle does not generate ATP; its function is to generate electrons in the form of NADH and $FADH_2$

- These reduced cofactors are those that in stage three of cellular respiration drive the synthesis of ATP

Production of Acetyl-coenzyme A

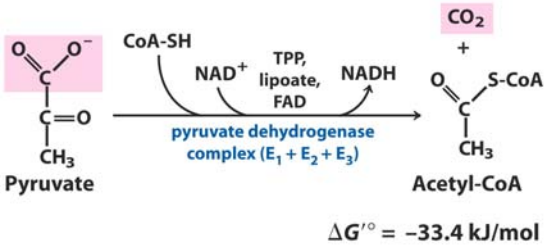
- As we have said, before entering the citric acid cycle, sugars and fatty acids must be converted to activated acetate units (acetyl-CoA.)
- **Fatty acids**, as we will see in another moment, have their own oxidation pathway to acetyl-CoA which is called **β-oxidation**
- Glucose, on the other hand, is initially oxidized to pyruvate (glycolysis) so that it is later oxidized and decarboxylated to acetate.

- The reaction is catalyzed by a multi-enzyme complex called pyruvate dehydrogenase (PDH) that is located in the mitochondria.
- The global reaction is an oxidative decarboxylation in which pyruvate loses its carboxyl group as CO2 and the two remaining carbons are transformed into an acetyl group attached to Coenzyme A (acetyl-CoA)

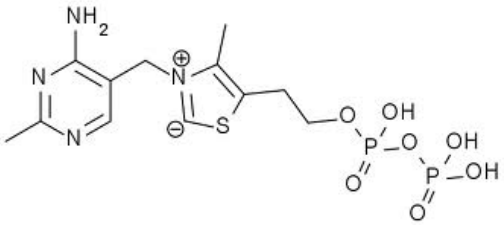
- In the total computation of the process, a pyruvate molecule is decarboxylated and oxidized to acetate bound to CoA and a NAD + molecule is reduced to NADH
- Therefore, the energy that is released during the decarboxylation and oxidation of pyruvate is transformed into a thioester bond with CoA and into a reduced electronic transporter.
- Later we will see how the subsequent oxidation of NADH is able to boost the synthesis of 2.5 ATP

The Pyruvate dehydrogenase complex (PDH)

- Under the name pyruvate dehydrogenase, three different enzymes and five coenzymes are agglutinated**: thiamine pyrophosphate (TPP), FAD, coenzyme A, lipoic acid and NAD$^+$**

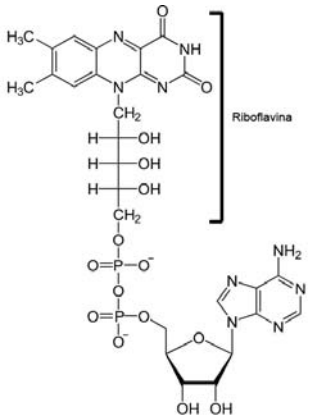


- It is made up of many copies of three enzymes:
E1: pyruvate dehydrogenase
E2: dihydrolipoyl transacetylase
E3: dihydrolipoyl dehydrogenase

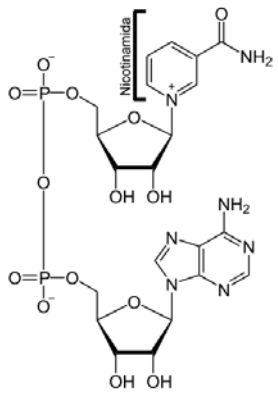


- **TPP**: is the coenzymatic form of vit B1. It participates in decarboxylation reactions. Its **deficit causes** a disease known as **Beriberi** that manifests itself with nervous and cardiac disorders. The absence of TPP prevents glucose from being oxidized, which has serious consequences on the nervous system
Exacerbated thiamine deficiency causes **Wernicke-Korsakoff syndrome**, characterized by memory loss, confusion, and paralysis. It is common among alcoholics due to poor intestinal absorption of thiamine

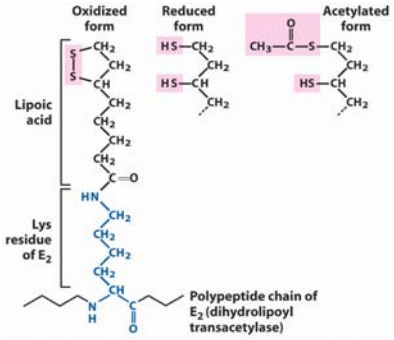
- **FAD** (flavin and adenine dinucleotide) is an electronic transporter formed by the union of two nucleotides, one with adenine and the other with riboflavin. Riboflavin is vitamin B2



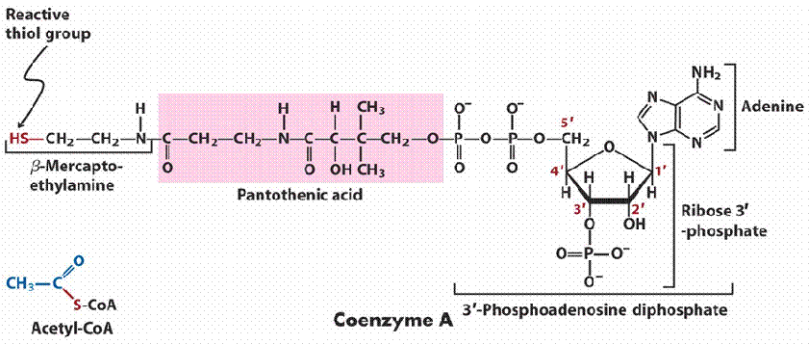
- **NAD$^+$** (nicotinamide and adenine dinucleotide) is an electronic carrier formed by the union of two nucleotides, one with nicotinamide (Vitamin B3) and the other with adenine. **Vitamin B3 deficiency causes Pellagra** characterized by rough, reddish skin (dermatitis), fatigue and encephalopathies



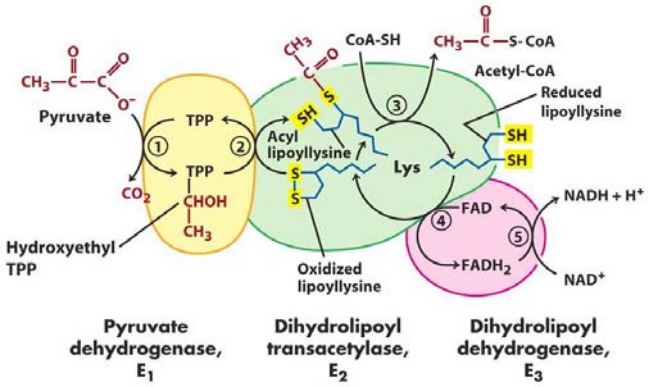
- **Lipoic acid**, derived from octadecanoic acid. It is characterized by two sulfhydryl groups that serve to bind the acetate derived from the decarboxylation of pyruvate and to transport electrons.
- The toxicity of As (arsenic) is due to the fact that it binds to the SH groups of lipoic acid, blocking PDH and other enzymes that contain this vitamin such as α-ketoglutarate dehydrogenase



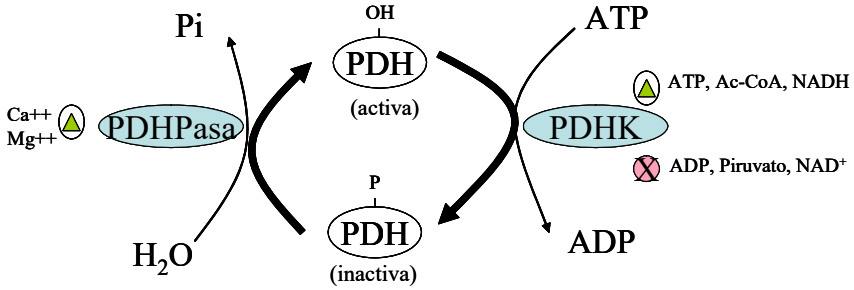
- **Coenzyme A** is a cofactor acting as a transporter and activator of the acetate derived from the decarboxylation of pyruvate. It is made up of three elements: **β-mercaptoethylamine, pantothenic acid and 3'-phosphoadenosine 5'-diphosphate**



- The reaction occurs in five stages such that the intermediate substrates never leave contact with the PDH complex, this process is **called substrate channeling**



- **PDH regulation**: there are two regulation mechanisms
  - **-allosteric inhibition** by NADH and acetyl-CoA. They both compete for binding sites on their enzymes and keep E2 in acetylated form, unable to accept the hydroxyethyl group of TPP.
  - **- covalent modification** by phosphorylation and dephosphorylation: dihydrolipoyl transacetylase has two enzymes linked:
    - **- PDH kinase**: inactivates PDH by phosphorylation in Ser; this kinase activity is stimulated by NADH and acetyl-CoA
    - **-PDH phosphatase**: reverses inhibition by dephosphorylation



Reactions of the Citric Acid Cycle

- The first difference that can be observed, after studying glycolysis, is that the latter does not regenerate glucose, is a linear sequence of reactions; on the other hand, Krebs cycle is a cyclic sequence of reactions: the first compound of the cycle, oxaloacetate, is regenerated at the end of the cycle
- At each lap of the cycle there is an **entry of one Ac-CoA and the release of 2 CO_2**

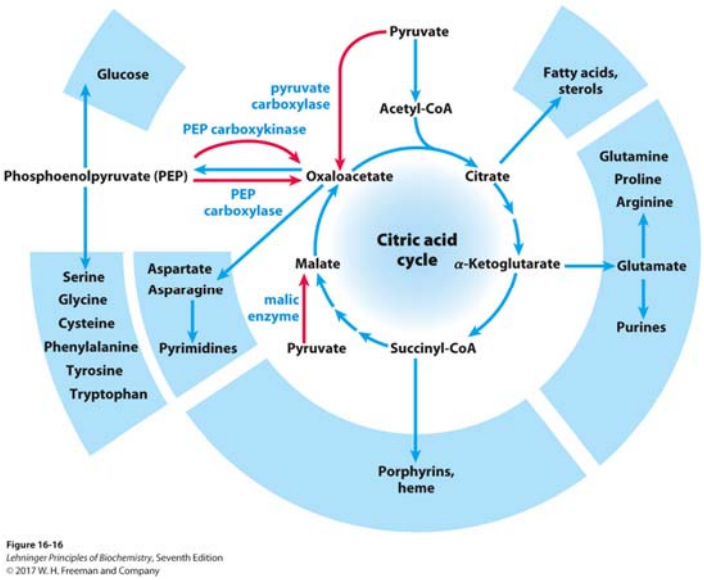


Figure 16-16
Lehninger Principles of Biochemistry, Seventh Edition
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- **Each turn uses an OA to condense with the Ac-CoA** to form citrate.
- In theory there is no loss of OA, however, some Krebs intermediates serve as biosynthetic precursors, thus the OA could be depleted preventing the entry of new Ac-CoA molecules.
- To avoid the loss of OA, cells have some reactions, called **anaplerotic**, to allow them to synthesize OA and prevent the cycle from stopping

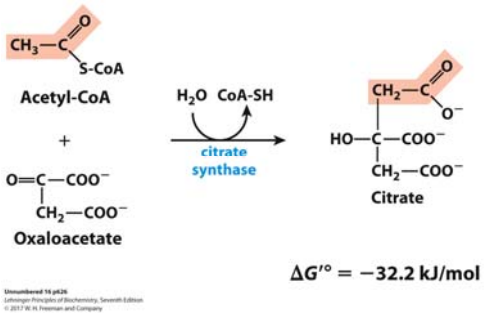
- We can say that Krebs cycle is not only a catabolic pathway but it is also an anabolic pathway. These types of pathways are called **amphibolic**.

- All reactions take place in the mitochondria; In addition, the enzymes necessary for the final transfer of electrons to O2 coupled with the synthesis of ATP (oxidative phosphorylation) are also in the mitochondria.

The eight steps of Krebs cycle

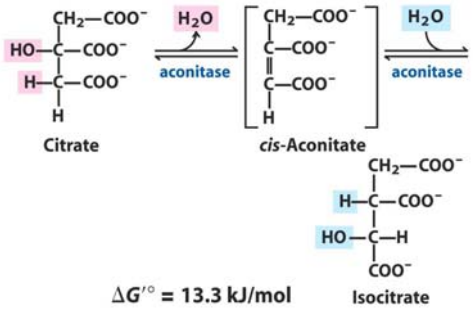
1. Formation of citrate

- The first reaction is the condensation of Ac-CoA with OA to form citrate
- It is catalyzed by citrate synthase. The methyl carbon of acetyl binds to the carbonyl of OA
- During the reaction, an intermediate compound, citroyl-CoA, bound to the active site of the enzyme, is generated. Hydrolysis of the thioester makes the reaction highly exergonic.
- In this reaction the CoA is recycled



2. Formation of isocitrate

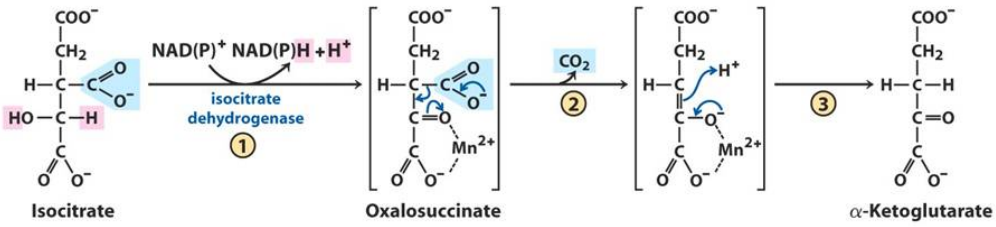
- Citrate is transformed to its isocitrate isomer
- Is a reversible isomerization reaction catalyzed by aconitase (aconitate hydratase)
- The reaction consists in the addition and elimination of a molecule of H2O
- Aconitase has a Fe-S center that serves to fix the substrate and to add or eliminate water.



- Although citrate is a symmetric molecule, aconitase acts only on the two carbons that come from OA.
- It is inhibited by ATP and NADH so that after a copious meal, there is an accumulation of citrate that will leave the mitochondria and will be used to synthesize fatty acids and therefore fats in the form of TAG
- Fluoroacetate is an aconitase inhibitor; it becomes fluoroacetyl-CoA which inhibits it.

3. Oxidation to isocitrate to α-ketoglutarate and CO_2.

- Isocitrate dehydrogenase catalyzes the oxidative decarboxylation of isocitrate giving rise to α-ketoglutarate and CO_2
-

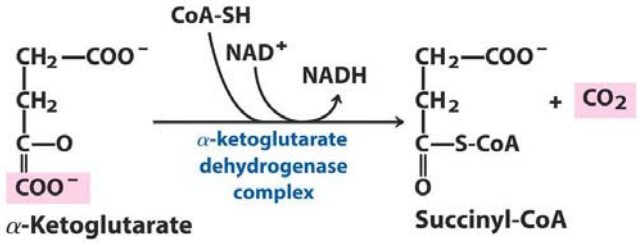


- There are **two isozymes of isocitrate dehydrogenase**: one of them is mitochondrial and uses NAD^+ as an oxidation-reduction cofactor, the other is both mitochondrial and cytosolic and uses $NADP^+$ as a cofactor.

- The $NADP^+$-dependent isozymes main mission is to generate NADPH necessary for biosynthetic reactions

4. Oxidation of α-ketoglutarate to succinyl-CoA and CO_2.

- Once again there is an oxidative decarboxylation catalyzed by the multienzymatic complex α-ketoglutarate dehydrogenase

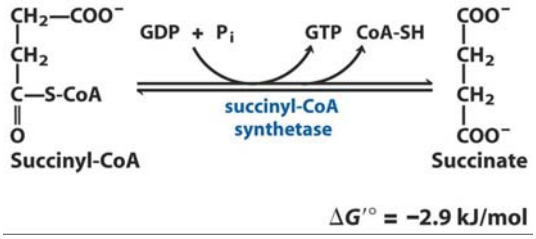


- The final electron acceptor is NAD^+
- **In this reaction, as occurred in PDH, the released energy is conserved as NADH and as a thioester bond in succinyl-CoA**
- Equally to the pyruvate dehydrogenase, **α-ketoglutarate dehydrogenase complex is made up of 3 enzymes and 5 coenzymes**
- The enzymes are:
  - E1: **α-ketoglutarate dehydrogenase**
  - E2: **dihydrolipoyl transuccinylase**
  - E3: **dihydrolipoyl dehydrogenase**

- The coenzymes are: **TPP, lipoic acid, CoA, FAD and NAD^+**
- The reaction is identical to that of PDH

5. Conversión del succinil-CoA en succinato

- In the next reaction, the hydrolysis of succinyl-CoA is coupled to the phosphorylation of GDP.
- **-The energy released at the hydrolysis of the thioester is used to promote the synthesis of a phosphoanhydride bond (GDP+Pi) and the formation of succinate and GTP**



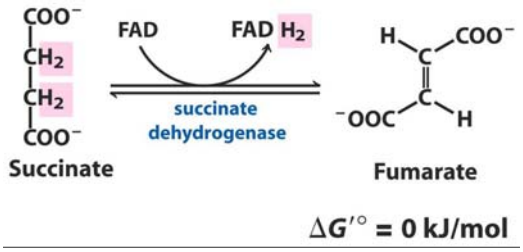
- The enzyme that catalyzes this reaction is **succinyl-CoA synthetase**
- The reaction takes place in 3 stages:
  1) Succinyl-CoA reacts with the Pi and forms succinyl-P and releasing CoA
  2) the P group of succinyl P is transferred to an His residue of the enzyme, releasing succinate
  3) the P group of His is transferred to GDP, forming GTP

- This type of mechanism in which ATP (or GTP) is synthesized is known as **substrate-level phosphorylations**
- The GTP produced in this reaction is easily converted to ATP through the action of nucleoside diphosphate kinase

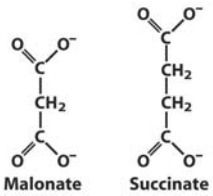
GTP + ADP ↔ GDP + ATP

6. Oxidation of succinate to fumarate

- The succinate formed from succinyl-CoA is oxidized to fumarate by the **succinate dehydrogenase**

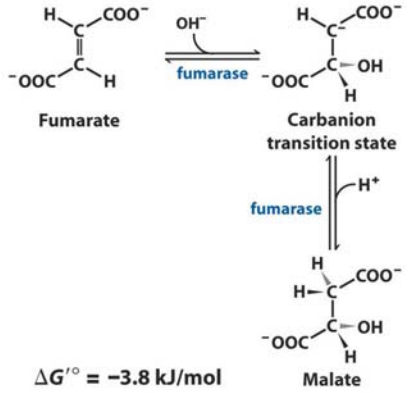


- This enzyme is a **flavoprotein (because contains FAD as cofactor); it is strongly attached to the inner mitochondrial membrane**.
- **The electrons** released during the oxidation of succinate **reduce FAD to $FADH_2$** and subsequently **pass to the mitochondrial electron transport chain where they will drive the synthesis of 1.5 ATP** molecules
- This reaction **is inhibited by malonate**, a structural analog of succinate. The addition of malonate to the mitochondria **blocks the Krebs cycle.**



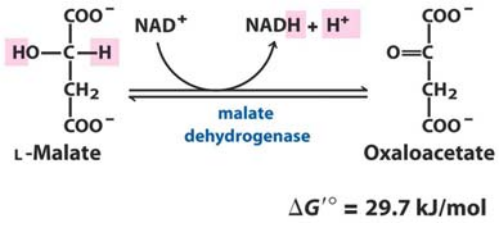
7. Hydration of fumarate a malate

- The **reversible hydration of fumarate to L-malate is catalyzed by fumarase.**
- **Fumarase is an stereospecific enzyme** and only can use the trans isomer of malate



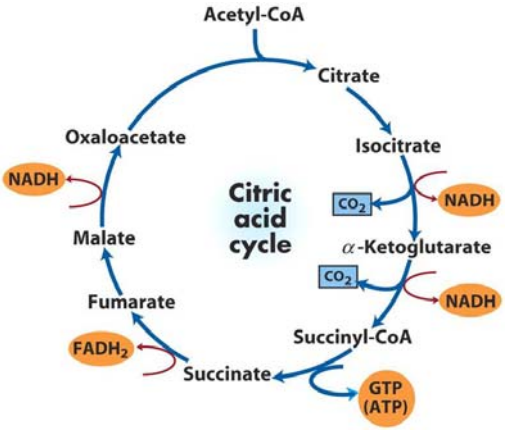
8. Oxidation of malate to oxaloacetate

- The last reaction of Krebs cycle consists on the **oxidation of malate** to oxaloacetate, catalyzed by **malate dehydrogenase**.
- Although the equilibrium of this reaction lies very far to the left, under physiological conditions, the continuous consumption of OA by citrate synthase makes the OA concentration low, so the reaction is shifted to the right..



Energy production

- Krebs cycle reactions lead to:
  1) **oxidation of an acetate group** (acetyl-coA) to **$2CO_2$**
  2) **3 NAD^+** are reduced to **3NADH**
  3) **1 FAD** is reduced to **1$FADH_2$**
  4) **1 ADP** is phosphorylated to **1ATP** (although in the cycle it is GTP, but we can assimilate it to ATP)



- Each molecule of NADH will lead the production of 2.5 ATP (oxidative phosphorylation)
- The molecule of $FADH_2$ will lead 1.5 ATP (oxidative phosphorylation)
- Therefore, each turn of the Krebs cycle generates 10 ATP

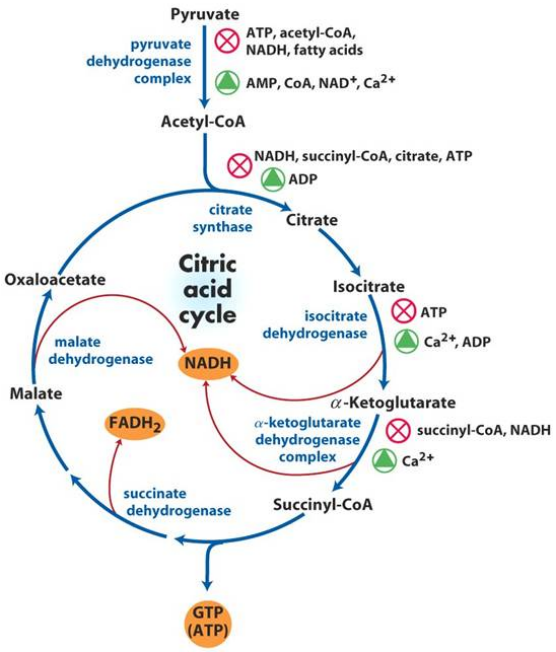
- If we consider the energy generated in the complete oxidation of a glucose molecule as a whole the balance will be:

Glucose →2 pyruvate + 2 ATP +2NADH (glycolysis)
2 pyruvate → 2 Ac-CoA + 2 CO_2+2NADH (Pyruvate dehydrogenase complex)
2 Ac-CoA → 4 CO_2 + 20 ATP (Krebs cycle)

- Making the corresponding conversions of NADH and $FADH_2$ to ATP (oxidative phosphorylation), we conclude that up to **32 ATP are obtained**

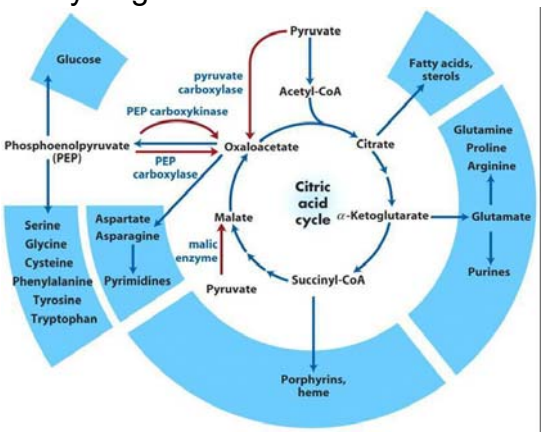
Regulation of the Krebs cycle

- The flow of carbon atoms from pyruvate to the Krebs cycle is subject to regulation at **two levels**:
  - **the conversion of pyruvate to acetyl-CoA**
  - **the entry of acetyl-CoA into the cycle**
- In other words, the enzymes that are subject to regulation are **fundamentally pyruvate dehydrogenase and citrate synthase.**
- It is also regulated at the level of isocitrate dehydrogenase and α-ketoglutarate dehydrogenase.
- We have previously seen the regulation of pyruvate dehydrogenase
- Regarding the reactions of Krebs cycle, there are three factors that govern the flow rate through the cycle:
  - **the availability of substrates**
  - **the inhibition by accumulated products**



- **the allosteric back-inhibition** of the enzymes that catalyze early stages of the cycle.

- In general, the overall activity of the cycle depends on the availability of NAD^+ for the oxidative reactions.
- Actually the availability of NAD^+ depends on the consumption of NADH by the respiratory chain, which in turn, depends on the rate of consumption of ATP by cells
- **The accumulation of NADH**, which occurs when ATP consumption is low, **inhibits citrate synthase, α-ketoglutarate dehydrogenase**. Under these circumstances, the malate dehydrogenase reaction is also reversed, which causes a decrease in the availability of oxaloacetate, thereby also slowing down the citrate synthase reaction.



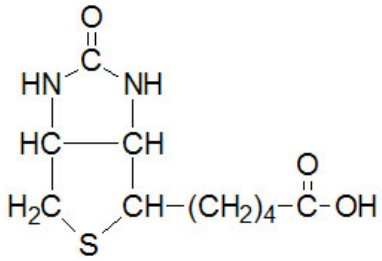
- **The accumulation of products** (citrate, succinyl CoA and ATP) inhibits the three limiting stages of the pathway: **citrate synthase, isocitrate dehydrogenase and α-ketoglutarate dehydrogenase)**
- **In muscle, Ca^{++},** which acts as a signal for contraction, **activates isocitrate dehydrogenase and α-ketoglutarate dehydrogenase, and, as we have seen previously, pyruvate dehydrogenase.**
- Citrate acts as an accommodator of the flow of metabolites between glycolysis and CAC since it inhibits phosphofructokinase I

Anaplerotic reaction to replenish Krebs cycle intermediates

- As we have already said, CAC provides precursors for many biosynthetic pathways.
- α-ketoglutarate and OA can serve as precursors of Asp and Glu, and through these compounds as nucleotide precursors.
- OA is a glucose precursor
- Succinyl-CoA is an intermediate in the synthesis of porphyrins
-
- To avoid a decrease in the concentration of OA as a consequence of the exit of intermediates from the cycle, there are some reactions that allow synthesizing OA or other intermediates; these reactions are called anaplerotic
- The most important anaplerotic reactions, in liver and kidney, is the reversible carboxylation of pyruvate to form OA, catalyzed by pyruvate carboxylase (liver and kidney)

Pyruvate + HCO_3^- + ATP ↔ OA + ADP + Pi

- This reaction consumes energy and requires the presence of biotin as a cofactor.
- Biotin is covalently linked to pyruvate carboxylase.



- Another element necessary in the reaction **is acetyl-CoA, which acts as an allosteric activator.**
- The accumulation of acetyl-CoA is fundamentally the consequence of a decrease in the concentration of OA.

- **In heart and skeletal muscle** there is another reaction in which Phosphoenolpyruvate (PEP) is converted into OA by phosphoenol pyruvate carboxykinase

PEP + CO_2 + GDP ↔ OA + GTP

- There is a third reaction consisting of the carboxylation of pyruvate to malate, catalyzed by the malic enzyme

Pyruvate + HCO_3^- + NAD(P)H ↔ Malate + $NAD(P)^+$

TABLE 16–2 Anaplerotic Reactions

Reaction	*Tissue(s)/organism(s)*
Pyruvate + HCO_3^- + ATP $\xrightleftharpoons{\text{pyruvate carboxylase}}$ oxaloacetate + ADP + P_i	Liver, kidney
Phosphoenolpyruvate + CO_2 + GDP $\xrightleftharpoons{\text{PEP carboxykinase}}$ oxaloacetate + GTP	Heart, skeletal muscle
Phosphoenolpyruvate + HCO_3^- $\xrightleftharpoons{\text{PEP carboxylase}}$ oxaloacetate + P_i	Higher plants, yeast, bacteria
Pyruvate + HCO_3^- + NAD(P)H $\xrightleftharpoons{\text{malic enzyme}}$ malate + $NAD(P)^+$	Widely distributed in eukaryotes and prokaryotes